AlpineWatch-1

Phase 0 Mission Proposal

Date: 2026-05-01
Version: MVP Draft

Executive Summary

During launch, the spacecraft is integrated with the selected launch vehicle, remains unpowered or in launch configuration as required by the launch provider, and separates into an initial low Earth orbit. During launch and early orbit operations, the spacecraft powers up, establishes a safe attitude and power-positive state, initializes command and telemetry links, and assesses basic spacecraft health. During orbit insertion or orbit acquisition, the mission uses the launcher-delivered orbit directly or performs any required orbit trim or acquisition activities needed to achieve the operational 550 km circular orbit at 97.6 degrees inclination, subject to the final propulsion concept. During commissioning, the mission team verifies bus subsystems, communications, power, thermal control, onboard storage, and attitude functions, then activates the multispectral payload, performs calibrations, and acquires initial test imagery to confirm imaging performance against mission requirements. During nominal operations, the spacecraft executes planned target imaging passes, stores payload data onboard between contacts, downlinks imagery and housekeeping telemetry during scheduled ground station passes, receives updated command loads, and repeats this cycle for the operational mission life. During contingency operations, fault detection logic or ground command places the spacecraft in safe mode, payload operations are suspended, essential power and thermal control are maintained, housekeeping telemetry is prioritized, and recovery actions are planned and executed from the ground. During end-of-life operations, routine imaging is terminated, residual energy sources are passivated as applicable, final telemetry is downlinked, and the spacecraft performs the approved disposal sequence or is placed in a compliant post-mission configuration consistent with the selected debris mitigation standard.

MVP Scope and Limitations

1. This MVP report is generated for a single-satellite mission concept. It does not analyze constellation deployment, multi-satellite phasing, revisit optimization, inter-satellite links, or multi-plane architectures.
2. This MVP report assumes an Earth Observation payload concept. Payload-specific analysis is limited to high-level swath, resolution, mass, and power assumptions where provided.

Mission Overview

- Acquire multispectral Earth observation imagery from a 550 km circular orbit at 97.6 degrees inclination with 5 m ground sampling distance over selected land targets.
- Provide imagery with a swath width of 40 km to support repeated monitoring of alpine and other terrestrial regions during a 3-year mission life.
- Downlink collected payload data and associated telemetry to ground stations for routine mission operations and image product generation.
- Demonstrate sustained on-orbit operation of a multispectral payload with 8 kg mass and 35 W nominal power consumption on an MVP Earth observation platform.

Mission Requirements

ID	Requirement	Rationale
MR-01	The spacecraft shall be compatible with injection into a 550 km c [illegible]	[illegible]
MR-02	The spacecraft shall support nominal on-orbit operations for not [illegible]	[illegible]
MR-03	The payload shall acquire multispectral imagery with a ground s [illegible]	[illegible]

ID	Requirement	Rationale
MR-04	The payload shall acquire multispectral imagery with a swath wi [illegible]	[illegible]
MR-05	The spacecraft electrical power subsystem shall provide not less [illegible]	[illegible]
MR-06	The spacecraft mechanical and structural design shall accommo [illegible]	[illegible] load mass.
MR-07	The attitude determination and control subsystem shall maintain [illegible]	[illegible]
MR-08	The spacecraft onboard data handling subsystem shall store pay [illegible]	[illegible]
MR-09	The communications subsystem shall downlink payload data an [illegible]	[illegible]
MR-10	The communications subsystem shall support command uplink d [illegible]	[illegible]
MR-11	The spacecraft shall complete a commissioning phase after orbit [illegible]	[illegible]
MR-12	The spacecraft shall implement an operational concept that inclu [illegible]	[illegible]
MR-13	The spacecraft fault management function shall command transi [illegible]	[illegible]
MR-14	The telemetry subsystem shall provide housekeeping telemetry f [illegible]	[illegible]
MR-15	The spacecraft shall execute an end-of-life disposal sequence th [illegible]	[illegible]

Mass Budget

Status: RED

Subsystem	Mass (kg)	Margin %	Notes

Power Budget

Status: YELLOW

Mode	Total W

Cost Estimate

Category	Cost (kEUR)	Basis
Spacecraft Hardware	7200.0	Engineering judgment
Integration & Test	2100.0	Engineering judgment
Launch	2800.0	Engineering judgment
Ground Segment	1400.0	Engineering judgment
Operations	1800.0	Engineering judgment
Program Management	1900.0	Engineering judgment
Contingency	4300.0	Engineering judgment

Component Alternatives

Assumptions and Open Items

Assumption: The launch vehicle and rideshare constraints will permit injection into an orbit that is either the operational 550 km circular orbit at 97.6 degrees inclination or an orbit reachable within the spacecraft delta-v budget.
Assumption: The payload optical design can achieve 5 m ground sampling distance and 40 km swath width from 550 km altitude within the spacecraft volume, mass, thermal, and pointing allocations.
Assumption: The spacecraft bus can accommodate an 8 kg payload and provide at least 35 W payload power during imaging while maintaining positive power balance over the full orbit.

Assumption: The planned ground segment will provide sufficient contact time and link performance to return the average daily payload data volume assumed for mission planning.
Assumption: The atmospheric drag environment at 550 km will not reduce orbit lifetime or increase maintenance delta-v beyond the propulsion or passive lifetime assumptions used for the mission concept.
Assumption: Radiation, thermal, and atomic oxygen environments for the selected orbit are within the qualification limits assumed for the spacecraft bus and payload components.
Assumption: Frequency allocation, licensing, and Earth observation regulatory approvals can be obtained within the project schedule and for the selected communications architecture.
Assumption: A debris mitigation compliance path exists for the mission, either through passive orbital decay within the allowed post-mission lifetime or through an active disposal capability.
Open question: What spectral bands, band centers, and bandwidths shall the payload provide, and what radiometric performance metrics shall apply to each band?
Open question: What maximum revisit time, target latitude range, and annual coverage fraction are required to define mission success for the selected land targets?
Open question: What image quality requirements shall be allocated, including MTF at Nyquist, maximum allowable image smear, signal-to-noise ratio, and geolocation accuracy?
Open question: What average and peak daily payload data volumes shall size onboard storage, downlink rate, and ground station contact time requirements?
Open question: What maximum interval between scheduled ground contacts shall the spacecraft be required to tolerate without loss of mission data or commanding capability?
Open question: Is onboard propulsion required, and if so, what delta-v shall be allocated for orbit acquisition, maintenance, collision avoidance, and end-of-life disposal?
Open question: What launch environment limits, separation interface requirements, and spacecraft mass and volume allocations shall be used before launch service procurement?
Open question: What commissioning and calibration concept shall be baselined, including duration, required calibration targets, and any onboard calibration hardware?
Open question: What autonomous fault detection, isolation, and recovery functions shall be implemented, and what faults shall trigger autonomous safe mode entry?
Open question: What debris mitigation standard and maximum post-mission orbital lifetime requirement shall govern the end-of-life design?

Disclaimer

This document is an AI-assisted Phase 0 concept draft. All technical values, cost estimates, procurement options, and requirements must be reviewed and validated by a qualified spacecraft systems engineer before use in design, procurement, proposal submission, or mission decision-making.